

Networks and Security

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VPN-

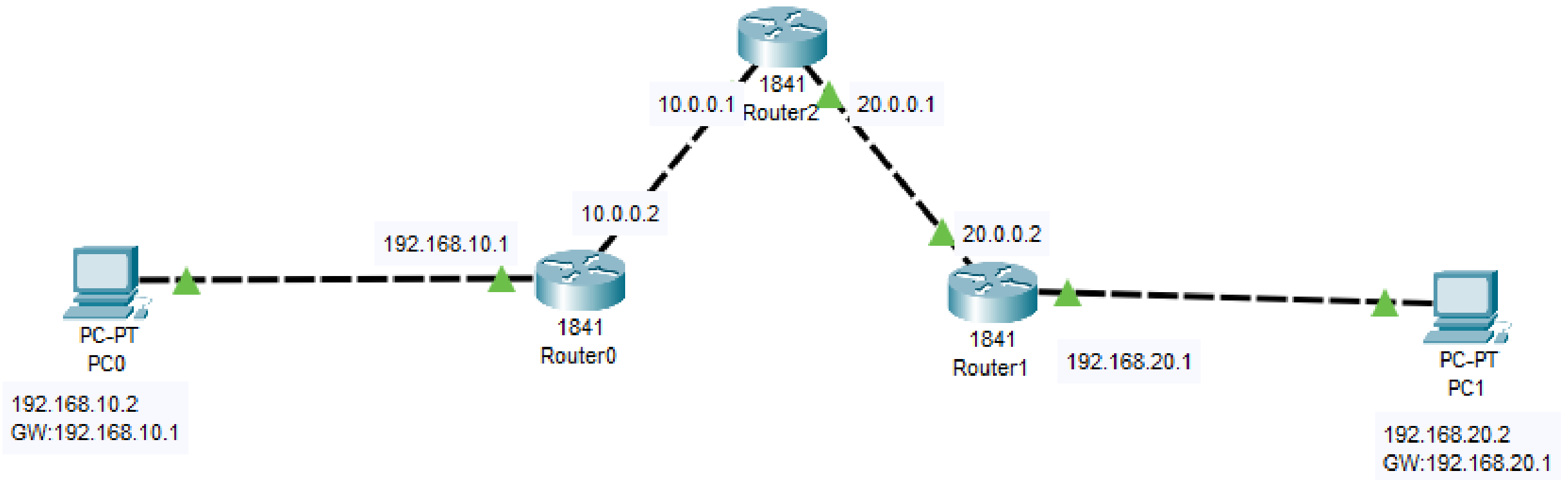
Generic Routing Encapsulation (GRE)
tunnel

Introduction

First, open the Cisco packet tracer desktop and select the devices given below:

Device	Model Name	Quantity
PC	PC	2
Router	1841	3

Topology



Apply RIP

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Network

Network Address
10.0.0.0
192.168.10.0

Router 0

Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Network

Network Address
20.0.0.0
192.168.20.0

Router 1

Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

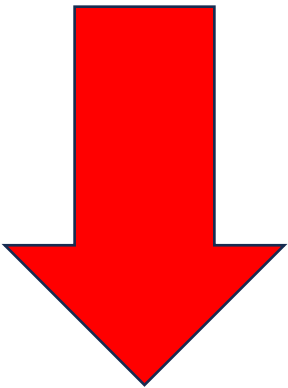
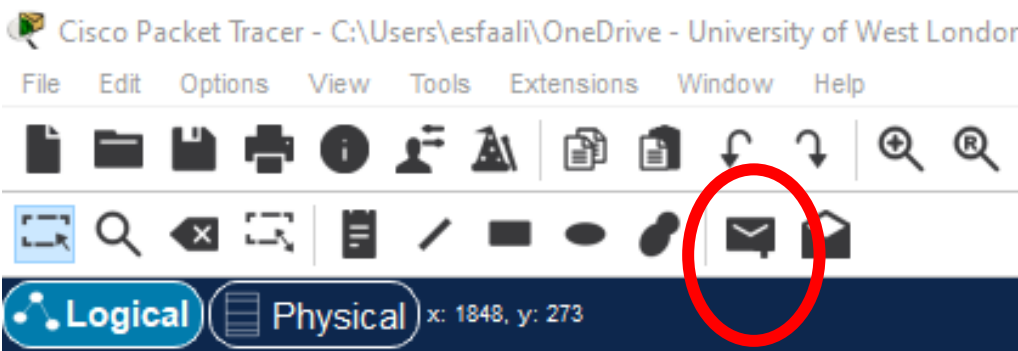
Network

Network Address
10.0.0.0
20.0.0.0

Router 2

Check the status

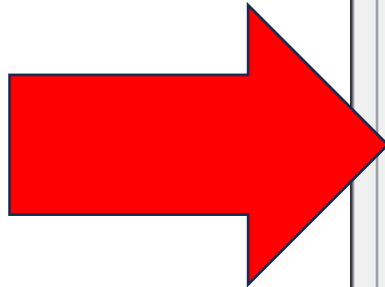
Click on  and drop it at PC0 or PC1 to see if they are connected!



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	D
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	

Check the status

Click on PC0 and go to Desktop, and then use ping and traceroute commands:



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=12ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>tracert 192.168.20.2

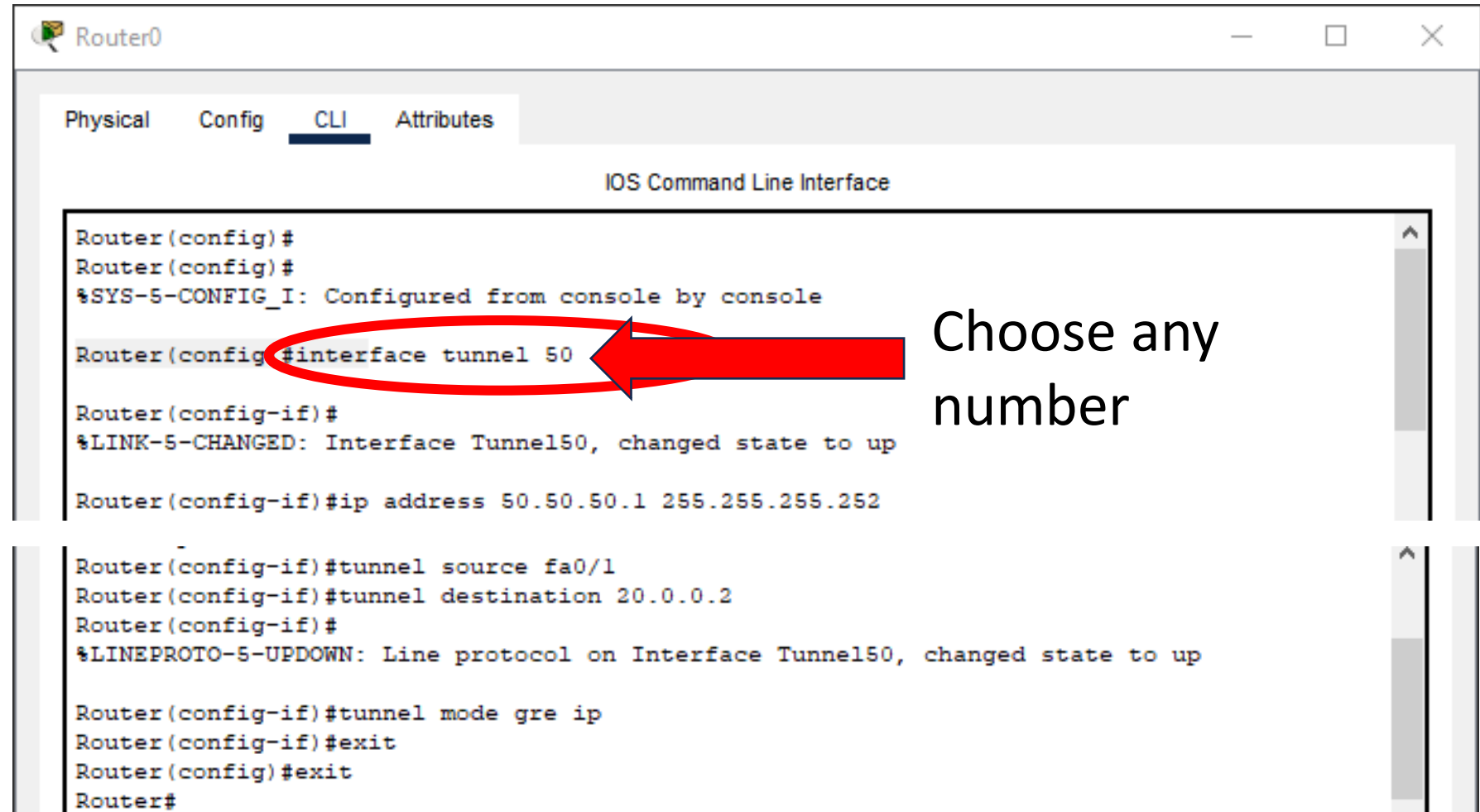
Tracing route to 192.168.20.2 over a maximum of 30 hops:
  0  0 ms    0 ms    0 ms    192.168.10.1
  1  0 ms    0 ms    0 ms    10.0.0.1
  2  0 ms    0 ms    0 ms    20.0.0.2
  3  0 ms    0 ms    0 ms    192.168.20.2
Trace complete.
```

Why? Ask, if it's not clear

Configure Tunnel for router 0

Tunnel set up

Follow the
screenshot line by
line



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config)#
Router(config)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config)#interface tunnel 50

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel50, changed state to up

Router(config-if)#ip address 50.50.50.1 255.255.255.252

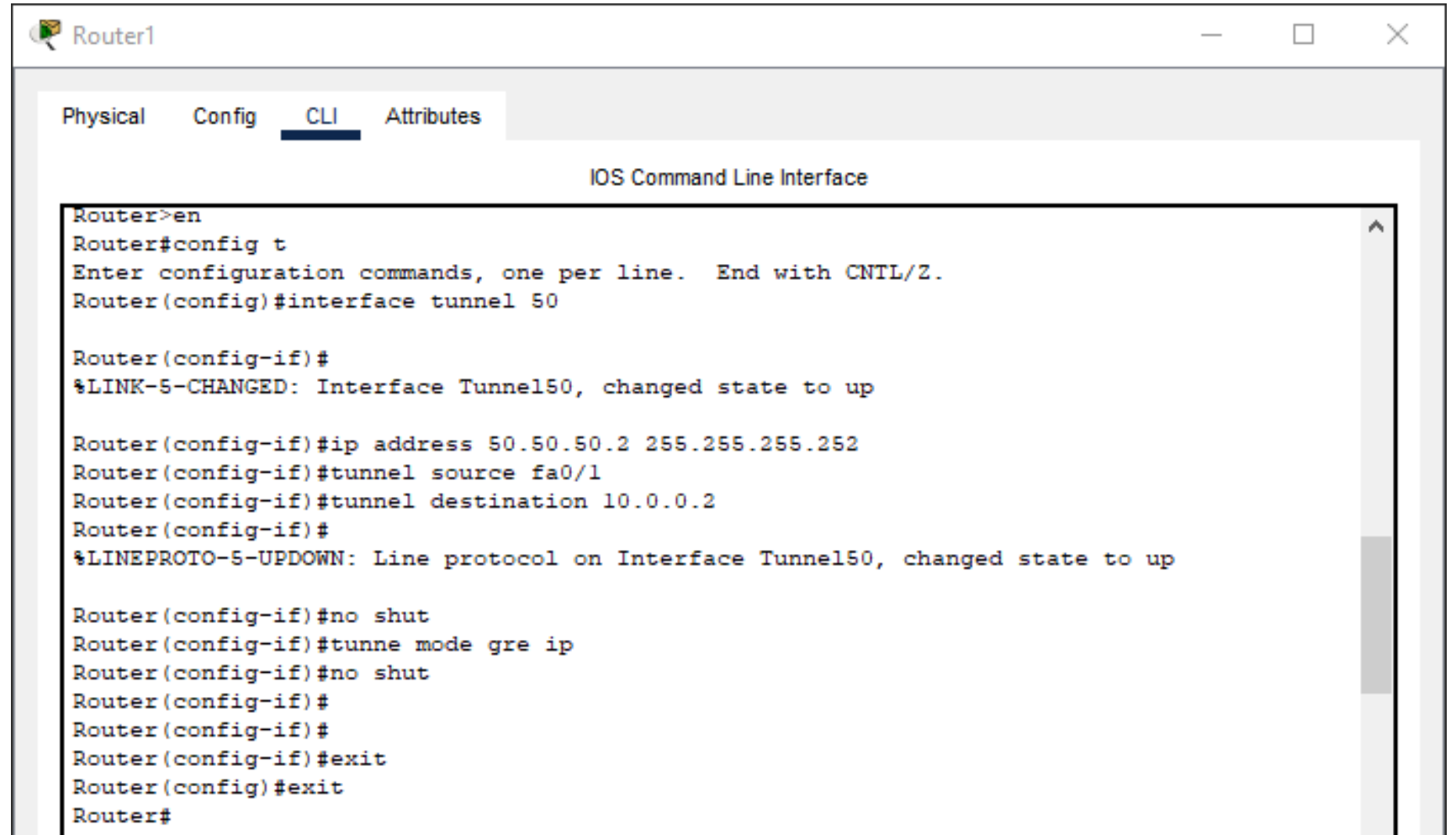
Router(config-if)#tunnel source fa0/1
Router(config-if)#tunnel destination 20.0.0.2
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel50, changed state to up

Router(config-if)#tunnel mode gre ip
Router(config-if)#exit
Router(config)#exit
Router#
```

Configure Tunnel for router 1

Tunnel set up

Follow the
screenshot line by
line



The screenshot shows a Cisco Router CLI window titled "Router1". The "CLI" tab is selected, and the "IOS Command Line Interface" is displayed. The following commands are entered:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface tunnel 50

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel50, changed state to up

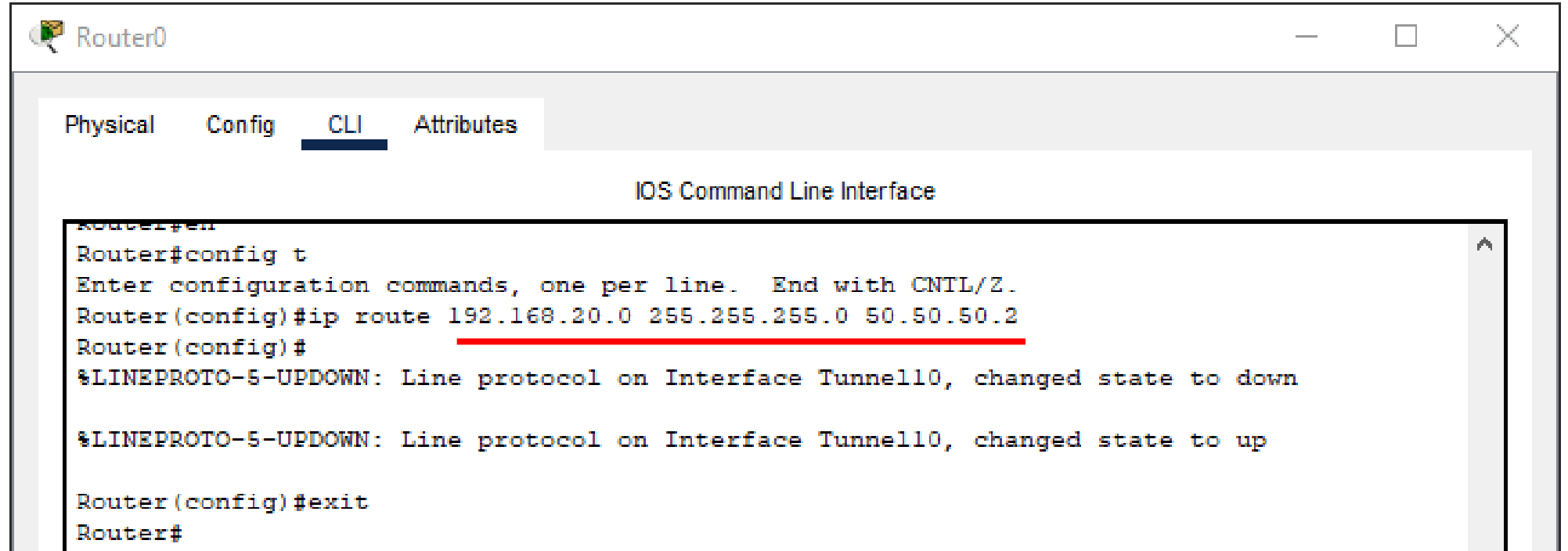
Router(config-if)#ip address 50.50.50.2 255.255.255.252
Router(config-if)#tunnel source fa0/1
Router(config-if)#tunnel destination 10.0.0.2
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel50, changed state to up

Router(config-if)#no shut
Router(config-if)#tunne mode gre ip
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
```

Configure routes for the Tunnel
between routers

Router 0

Follow the
screenshot line by
line

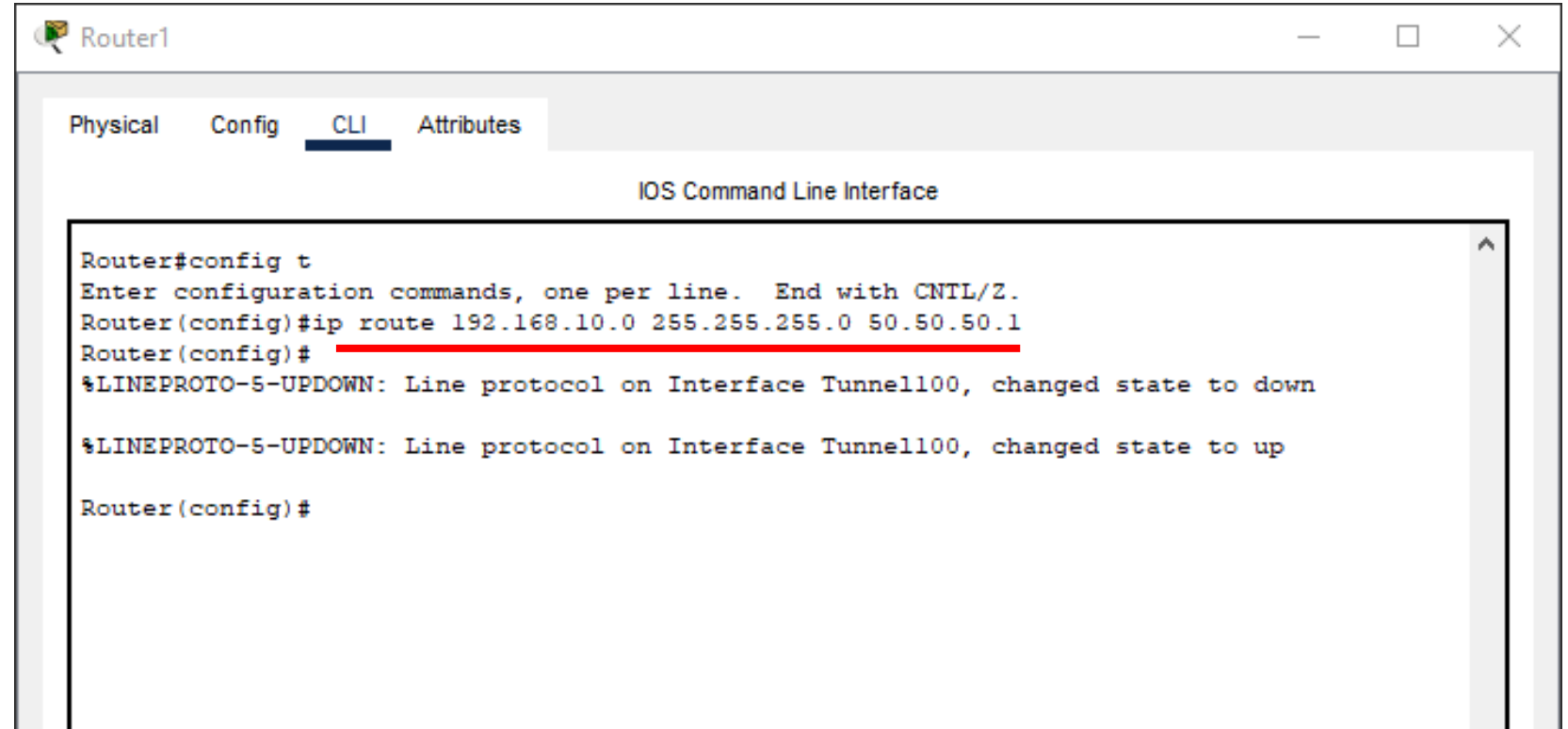


The screenshot shows a window titled "Router0" with a tabbed interface. The "CLI" tab is selected, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
Router#  
Router#config t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#ip route 192.168.20.0 255.255.255.0 50.50.50.2  
Router(config)#  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel10, changed state to down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel10, changed state to up  
  
Router(config)#exit  
Router#
```

Router 1

Follow the
screenshot line by
line

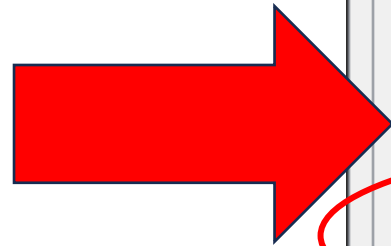


The screenshot shows a window titled "Router1" with a tabbed interface. The "CLI" tab is selected, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.10.0 255.255.255.0 50.50.50.1
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel100, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel100, changed state to up
Router(config)#
```

The command `ip route 192.168.10.0 255.255.255.0 50.50.50.1` is highlighted with a red underline in the original image.

Check the status



This is what you need to show us

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=12ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>tracert 192.168.20.2

Tracing route to 192.168.20.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.10.1
  1  0 ms    0 ms    0 ms    10.0.0.1
  2  0 ms    0 ms    0 ms    20.0.0.2
  3  0 ms    0 ms    0 ms    192.168.20.2

Trace complete.

C:\>tracert 192.168.20.2

Tracing route to 192.168.20.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.10.1
  1  *        *        1 ms    50.50.50.2
  2  0 ms    0 ms    0 ms    192.168.20.2

Trace complete.

C:\>
```

Top 15

Topology

This is what we created

